Protecting Salem's Ecosystem:

Predictive Modelling of Tree Canopies and Stream Temperatures

Siera Edwards, Serenna Walter, and Brooke Proctor

Team Metadata

Team Members

Serenna Walter Owner Product Lead	Siera Edwards	Brooke Proctor
slthorsen@willamette.edu	smedwards@willamette.edu	blproctor@willamette.edu

Peer Stakeholder POs: Spencer Fiden and Amaya Supancich-McCord

Studio Session: 3

Team Size: 3 people

Problem Framing and Research Question

Bodies of water around the world are beginning to heat up at faster rates than air heat waves (Bush, 2025). These water heatwaves–defined by at least five consecutive days above the seasonal average–result in lower dissolved oxygen content, nutrient cycling, and pH buffering (PennState, 2026). All of this means that ecosystems and native species adapted to live in colder waters face serious threats.

One way to combat high stream temperatures is to support riparian zones, the dense plant areas along rivers (University of Georgia, 2026). These areas often sport trees and other foliage that shade and cool rivers and streams, effectively lowering the water temperature. However, these areas are at risk for several reasons.

The first risk to riparian zones is human development built alongside waterways (University of Georgia, 2026). If projects such as Salem's proposed Riverfront Redevelopment Site are not done explicitly with the ecosystem in mind, there can be negative consequences due to deforestation and disruptions to the riparian zones.

Secondly, Emerald Ash Borers are a new concern in Salem. Emerald Ash Borers are pests that are native to Northern China, Russia, Japan, and Korea (USDA, 2023). These insects are highly corrosive to North American Ash Trees and are killing trees across the United States (Penn State Extension, 2025). These beetles were found in Oregon for the first time in 2022 and have been steadily increasing their lifespan as the climate changes (Boboc, 2026). This recent expansion caused Oregon to start thinking critically about readiness and response plans, including Salem's own plan (City of Salem, 2026).

With these complex system problems in mind, four research questions were developed:

1. Can we use historic stream temperature data to predict stream temperature for 2020, 2022, and 2024?

2. Can we combine historic stream and weather data to get a more accurate prediction of stream temperatures in 2020, 2022, and 2024?
3. Can we combine historic stream and canopy data to get a more accurate prediction of stream temperatures in 2020, 2022, and 2024?
4. Will combining all three historic datasets–streams, weather, and canopy–produce a highly accurate prediction of stream temperature in 2020, 2022, and 2024?

Our vision for this project is to help government and private organizations prevent the loss or disruption of riparian zones in Salem. By developing an interactive map and dashboard that models the potential effects on Salem's waterways, this project will identify at-risk areas and highlight targeted preventative measures that can be taken. There is no existing research on this specific area that we are aware of.

We are specifically interested in this project because the three of us grew up in the Willamette Valley, surrounded by nature and a curiosity for stewardship. This unification brings a personal element to the heart of the project.

Impact and Stakeholders

The results of this project will benefit the City of Salem's Department of Natural Resources, which is a stakeholder of the project. Using these results, a more informed plan can be created by the City of Salem to protect riparian zones in Salem, along with the indigenous species that live in them.

The Shared Theme domain for our project is the Pacific Northwest forest and environmental health. Our Peer POs are looking into fire patterns in Oregon and could support different perspectives about topics such as tree weaknesses from drought or heat, as well as heat ecology damages in general that may compound our data. This will help us think holistically about the project.

Data Sources and Access Plan

Data Source	Brief Description	Access	Volume/Cadence	Instructor Approval Status
Tree Plotter Canopy [1]	Very similar to above. Maps tree types and their locations. Only in the	Public data can be scraped or downloaded	As published, estimated less than or equal to once a year.	Requested

	Salem area. Publicly Available.	using a stakeholder's login information		
Stream/ Waterway from Dr. Gore [2]	This data contains two different CSVs, one with 15-minute interval measures for two different sites and the other with daily measurements for two different sites. Data was collected on the Mill Creek.	Accessible via request from Dr. Gore at Willamette University.	Daily. We may aggregate this to the weekly level. Data was collected from 2014-12-31 to 2025-6-4.	Requested
Salem Weather Data [3]	This data is one CSV with daily temperature and weather summaries for Salem, collected at McNary Field	Downloadable via request form from NOAA	Daily, with information from 1900-01-01 to 2026-06-27	Approved

Links for datasets

1. https://pg-cloud.com/SalemOR/
2. https://sites.google.com/willamette.edu/castaway-club/mill-creek-research-project
3. https://www.ncdc.noaa.gov/cdo-web/cart

Data Engineering Plan

Ingestion: The data will be ingested by downloading the data as flat CSV files.

Stored: The data will be stored in the data/raw folder on the project repo. Additionally, the data is accessible to all researchers via a Railway database.

Versioned:

- data/raw/places/: download CSV snapshots versioned by download date.
- data/interim/: cleaned data (with row counts, missingness, and duplicate checks), but not engineered.

- data/processed/: analysis-ready panel for modeling.

Documented: Our data will be documented each time it is uploaded with the name, details of ingestion, and what changes or cleaning were done.

After ingestion, the data will be cleaned using R. This RMarkdown will be included in the data/interim/ folder. The cleaned data will then be migrated into a PostgreSQL database hosted by Railway so that each of our team members has access to all the data.

Planned Analysis

1. **Set up Railway:**
    a. **Create:** Railway Project for our capstone. This will allow us to ingest data from CSV files into a shared space and set up web scrapers. Also, set up a connection to this project in our individual PgAdmin or Beekeeper.
    b. **Observe**: N/A
    c. **Analyze**: N/A
    d. Success: We will have a functional Railway Project and successfully connect to it from each of our devices.
2. **Collect the data:**
    a. Create: We will download and scrape all of the data sources using an API.
    b. Observe: We will ensure data has been populated into the table and note the data structure for each.
    c. Analyze: We will look into the raw CSVs to look for obvious errors, such as duplicate rows or discrepancies

   d. **Success:** We have at least one data source on Emerald Ash Borer Prevalence, at least one on Tree types and locations, at least one on Oregon Ash Borer Trap Data with recurrence via API, and successfully collected the Stream/Waterway Data.

3. **Exploratory Data Analysis:**
   a. **Create:** We will create basic summaries of each dataset with descriptive statistics included.
   b. **Observe:** We will consider whether these summaries are relevant for inclusion in the final report.
   c. **Analyze:** We will check in with our peer POs for analysis of the data to confirm that the statistics are sensible.
   d. **Success:** We collect the mean, median, mode, and quartiles for every numeric column. We collect all possible values for qualitative columns, as well as the most common value for each column.

4. **Regression model for stream temperature (RQ1):**
   a. **Create:** We want to create a model that can predict stream temperatures for (stream names) for the next 3 years using historic data. We will train the model on data from 2020, 2022, and 2024. We will create a Linear Regression, Polynomial Regression, Random Forest Regression, Support Vector Regression, ElasticNet Regression, and Bayesian Linear Regression model to find the best fit.
   b. **Observe:** We will pursue and fine-tune the model with the highest accuracy and cross-validation score.
   c. **Analyze:** We will evaluate the models based on R-squared, MAE, RMSE, and the cross-validation score.

d. **Success:** Achieved a model with a relatively low RMSE and MAE, an R-squared above 75%, and a cross-validation score above 75%.

5. **Regression model for streams with the addition of weather data (RQ2):**
   a. **Create:** We want to create a linear regression model including streams and weather data that can predict stream temperature across 2020, 2022, and 2024. We will model this using the few highest accuracy models from RQ1.
   b. **Observe:** We will pursue and fine-tune the model with the highest accuracy and cross-validation score.
   c. **Analyze:** We will evaluate the models based on R-squared, MAE, RMSE, and the cross-validation score.
   d. **Success:** Achieved a model with a relatively low RMSE and MAE, an R-squared above 75%, and a cross-validation score above 75%.
6. **Regression model for streams with the addition of tree canopy (RQ3):**
   a. **Create:** We want to create a linear regression model including streams and tree canopies that can predict stream temperature across 2020, 2022, and 2024. We will model this using the few highest accuracy models from RQ1.
   b. **Observe:** We will pursue and fine-tune the model with the lowest RMSE, the lowest MAE, the highest R-squared, and the highest cross-validation score. value.
   c. **Analyze:** We will use the last 2 years of data for the test set and the previous data for training. We will evaluate the model based on RMSE, MAE, and R-squared value for predicting values for the next 5 years.
   d. **Success:** Achieved a model with a relatively low RMSE and MAE, and an R-squared above 75%

7. **Regression Model for Stream Temp Using Extrapolated Canopy Coverage and Weather Data (RQ#4)**
    a. Create: Based on our findings from RW#2 and RQ#3, we will extrapolate the data and fill in values for future weather and canopy coverage. We will use these values as additional predictors to create another stream temperature model. We will train the model on data from 2018 to 2023 and test it on the data from 2024 to 2026. Then, we will use the model to predict further until 2029. We will create a Linear Regression, Polynomial Regression, Random Forest Regression, Support Vector Regression, ElasticNet Regression, and Bayesian Linear Regression model to find the best fit.
    b. Observe: We will fine-tune the model to achieve the lowest RMSE, the lowest MAE, the highest R-squared, and the highest cross-validation score.
    c. Analyze: We will evaluate the models based on R-squared, MAE, RMSE, and the cross-validation score.
    d. Success: Achieved a model with a relatively low RMSE and MAE, an R-squared above 75%, and a cross-validation score above 75%. Ideally, all of these values would be better than the model created in RQ#1.

Program Integration

We intend to use our knowledge in research design to guide our project plan, create and guide our data collection and data analysis methods. We have used these skills in previous classes, which have given us an understanding of how to safely collect public data and load it into a database. We have also had a lot of practice performing EDA, so we can leverage this experience to clean and analyze our data effectively. We will use our data engineering skills to bring all this data together in a single database (Postgres). Once this data is brought together and cleaned, we will be able to use our knowledge from the Machine Learning courses to test different models on our data to answer our research questions. We will also use general data summaries and analysis to explain our data and the context around it.

Our background in the Data Engineering course will also support our goal of creating a Grafana Dashboard to summarize our stream temp predictions and general data summaries. For any graphs used in the Grafana Dashboard, we will leverage our teachings from Data Visualization to create easy-to-read and informative visualizations. Finally, we will continue to use our skills from Data Visualization when creating an interactive map to demonstrate the areas where streams are most at risk of rising temperatures and where tree prevalence could see the greatest impacts.

Ethics and Risk Notes

It is always possible that our models are not accurate enough, which could lead to protective measures being taken in the wrong areas while areas in need of protection are left at risk. For example, a faulty model, while crafted to the best of our abilities with the data that we have, may predict the wrong areas that would be at risk for stream temperature rises, leading to protection in the wrong areas. We could also potentially underestimate or overestimate the risk of stream temperature changes. Additionally, we cannot predict the random removal of trees due to construction or other human-caused events. To address these issues, we will clearly state the limitations of our data’s time range, things that we cannot predict in our data, and the nature of using extrapolation in our modeling. We do not anticipate any issues with data accessibility since we are using open data, as we have consent from the City of Salem to use this information publicly.

We anticipate our main risk for this project being a lack of time. We worry that gathering and collecting this data will be time-consuming, and we want to ensure that we are thorough to avoid little mistakes early on. To mitigate this, we will generally try to start tasks early enough that we can ask for help or fix errors early should they arise. Another potential risk is that we don't have enough data to create a successful model. We feel that this is likely not going to be a problem, but we won't know for sure until we do some EDA on the data and begin that process. An additional risk we are concerned about is the usability of a few of our data sources. Some data sources may not be reliable or have large data quality issues, which may require us to pivot to a different source. To mitigate these data quality risks, we have planned backup data sources for these datasets that we anticipate may have issues.

Timeline and Milestones

Week 4: Project Proposal Completed, Data Sources Identified

- We made it through these steps before writing this document and didn't run into any problems.

Week 5: Download all data, ingest it into Postgres, and start Railway

- Risks: Can we download the data? Is the data usable? Do we have access to the data? Do we have to use web scrapers to gather the data? Do the APIs work effectively? Can we remember how to start a Railway Project?
- Mitigation: Start early! We want to start exploring these as soon as possible so that we have time to deal with all these potential risks. Most of the data we have gotten from

people working for the State of Oregon, so we feel confident that we can make it work, but we won't know until we try.

Week 6: Data Cleaning Complete and Final Data Sources Selected

- Risks: Do we understand the values in the data? Do our data sources allow us to answer our research questions?
- Mitigation: We will continue to reach out to our connections at the State of Oregon to understand the data if needed. And we will continue to stay ahead on this project so that we can continue to search for new data sources if we don't feel like the current ones can answer the research question. Make sure we have enough variables to train models on the data.

Week 7: Data Summary Completed and EDA/Feature Engineering

- Risks: We can't engineer any features. We are tight on time and won't get all the ingestion done on time.
- Mitigation: We follow this plan and really push to get the milestone done on time. We should always be able to engineer new features from the data if we understand it and the context of our subject well. If we keep the modeling in mind while bringing in the data it should keep us on track.

Week 8: Amend Project Proposal, Data Summary, EDA, Start Models for RQ#1, #2, #3 (Document process as we go)

- Risks: We don't get reliable data from our sources. We run short on time and fall behind schedule.

- Mitigation: We stay in close contact with our sources to stay updated on their ability to send us the new data we need. We work on this project in any spare time we have and utilise our expert feedback in the class meetings.

Week 9: Final models for RQ#1, #2, #3. Final model for RQ #4. Draft Complete of Interactive Map and Dashboard Design

- Risks: Any complications with the predictive models. The interactive map could be very challenging or time-consuming to make. We might need multiple interactive maps to show data over time, which might be clunky.
- Mitigation: Reach out to Prof. Smalley if we need help with aspects of data visualization. Ask for help in general if we get stuck. Make sure we start these aspects early and create a plan in week 8 so we only have to implement it this week.

Week 10: Poster Rough Draft

- Risks: We could potentially not have enough information to answer our research questions to create a good draft.
- Mitigation: In the early stages of our modeling, we can have a sense of whether we will have enough information or not to create a comprehensive draft to answer our questions. We can explain where things went wrong, and if necessary, we could grab emergency data to make our project meaningful again.

Week 11: Finalize Map/Dashboard and Poster Setup, Begin Write-Up

- Risks: We could realize that our data doesn't actually support our research question or isn't actually answering the questions posed to us by our partners at the State or Oregon.
- Mitigation: Be able to explain why things went wrong. If really necessary, we could go back and gather emergency data to make our project meaningful again.

Week 12: Complete Write-Up Draft

- Risks: We don't have the project pieces finished, so we can't complete parts of the write-up.
- Mitigation: Any draft is still better than none. We could write about what we assume will happen when we finish those missing sections and tidy it up/ correct it for the final submission.

Week 13: Final Edit of Write-Up and Practice Final Presentation

- Risks: Our write-up draft is not complete yet, and it will take extra time to make it complete before editing. Our final presentation relies on our draft being complete so that we can effectively communicate our findings.
- Mitigation: We will make sure we are on time with the write-up so that we are not pressed for time in the editing stages, and when we need to create our presentation.

Week 14: Final Presentation

- Risks: Someone is ill. We generally don't feel prepared.
- Mitigation: We all still know each other's parts of the project so we aren't scrambling the morning of the presentation. We should be practicing the presentation ahead of time so that we know what to expect on presentation day.

DS3 Process and Tooling

- GitHub repo: https://github.com/Serenna4/Emerald-Ash-Borer-Capstone-Project
- GitHub Projects board: https://github.com/users/Serenna4/projects/2
- All team members have joined Discord and GitHub. We have also gained access to the projects for which we are Peer POs.

- The CHARTER.md document is completed, and a backlog is created.
- We have agreed to have our Studio Briefs due Sunday at 5 pm, so they are ready for the next day's class. The critiques are due Wednesday at midnight. And updated Backlogs / Iteration Review should be completed by Saturday at 5 pm so that the Briefs can be written on time.

References

Boboc, J. (2026). *Emerald Ash Borer pests are emerging in Oregon at higher rates in May*. KLCC. https://www.klcc.org/environment/2026-05-25/emerald-ash-borer-pests-emerging-in-oregon-at-higher-rates-in-may

Bush, E. (2025). *U.S. rivers are experiencing unprecedented and unexpectedly intense warming.* NBC News. https://www.nbcnews.com/science/climate-change/us-rivers-are-experiencing-unprecedented-unexpectedly-intense-warming-rcna233002

City of Salem. (2026). *City of Salem Emerald Ash Borer Readiness and Response Plan*. https://www.cityofsalem.net/home/showpublisheddocument/30563/639144417587730000

Hudgins, E. et al. (2024). *Spread management priorities to limit emerald ash borer (Agrilus planipennis) impacts on United States street trees*. Conservation Science and Practice. https://conbio.onlinelibrary.wiley.com/doi/full/10.1111/csp2.13087

Huset, R. (2013). *A GIS-based analysis of the environmental predictors of dispersal of the emerald ash borer in New York.* Syracuse University. https://www.proquest.com/openview/a62b54ffa5c9c275cbee2644dfecd010/1?pq-origsite=gscholar&cbl=18750

Mercker, D. and Hoyt, K. (N.D.) *Forest Stand Delineation: How Is It Done and Why?* University of Tennessee Extension Institute of Agriculture. https://utia.tennessee.edu/publications/wp-content/uploads/sites/269/2023/10/W982.pdf

Oregon Department of Forestry. (2024). *Emerald Ash Borer (EAB) Agrilus planipennis Fairmaire*. https://www.oregon.gov/odf/Documents/forestbenefits/fact-sheet-emerald-ash-borer.pdf

Oregon State University. (n.d.) *Riparian Areas - Stream Side Protection.* Oregon State University. https://agsci.oregonstate.edu/water-quality-management/riparian-areas-stream-side-protection#:~:text=Riparian%20zones%20or%20areas%20have,from%20those%20of%20adjacent%20uplands

Penn State Extension. (2026). *We are in Hot Water – Streams and Rivers are Heating Up.* Pennsylvania State University. https://extension.psu.edu/we-are-in-hot-water-streams-and-rivers-are-heating-up

Penn State Extension. (2025). *Emerald Ash Borer*. Pennsylvania State University. https://extension.psu.edu/emerald-ash-borer

University of Georgia. (2026). *Stream Warming and Brook Trout in the Southeast*. University of Georgia. https://storymaps.arcgis.com/stories/834fb9f543424c96ab3f37b677d710fa